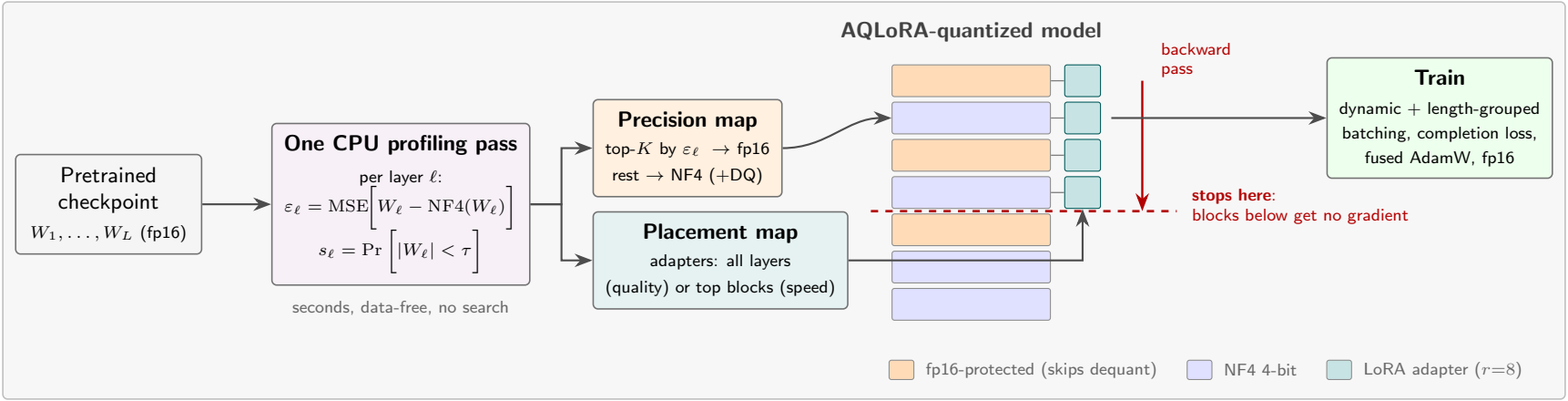

Pretrained checkpoint
$W_1, \ldots, W_L$ (fp16)
One CPU profiling pass
per layer ℓ:
$\varepsilon_\ell = \mathrm{MSE}\left[W_\ell - \mathrm{NF4}(W_\ell)\right]$
$s_\ell = \Pr\left[|W_\ell| < \tau\right]$
seconds, data-free, no search
Precision map
top-K by ε_ℓ → fp16
rest → NF4 (+DQ)
Placement map
adapters: all layers
(quality) or top blocks (speed)
AQLoRA-quantized model
backward pass
stops here:
blocks below get no gradient
Train
dynamic + length-grouped batching, completion loss, fused AdamW, fp16
fp16-protected (skips dequant)
NF4 4-bit
LoRA adapter (r=8)